

User Manual for Coil Stacks in Marylie

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1 Abstract

The coil stack is a super-element option for Marylie consisting of a collection of soft-edge straight-axis cylindrical magnets on a straight axis. The term “soft-edge” refers to the fact that the magnetic field and on-axis gradient of the magnets extend smoothly beyond the physical ends of the magnet. On-axis generalized gradients and their z derivatives at arbitrary z points up to the required order are computed by superposition of gradients of all of the individual magnets in the stack. The superimposed on-axis gradients are used to compute the transfer map through the whole coil stack by numerical integration in z . Coil-stack elements are identified by the keyword `coil`.

2 Coil element format in Marylie input file

Marylie input files are frequently identified with the suffix `.mip` and therefore will be called mip files in this manual. Coil element blocks in a mip file have a fixed format starting with two character strings in the first line followed by 11 floating-point numbers in the second through fifth lines. The first character string is a user-supplied coil element name of up to 6 characters. The second character string in the first line is always `coil`. The meaning of the 11 numbers is dependent on the coil type. If all of the 11 numbers are not needed for a particular coil type, the user sets unneeded numbers to zero.

Coil stacks must contain only coil-type elements and are defined by a `#lines` command in the mip file (this will become more clear in the examples in a following section). Coil elements in a coil stack are either magnet elements or spacer elements. A third type of coil element is the coil stack initialization element. The coil stack initialization element is not part of a coil stack, but is one of the commands in the `#labor` part of the mip file and precedes the user-supplied coil stack name in the `#labor` command list. The first character string in the first line of the initialization element is user-supplied and is usually chosen to be `cinit` and, as always, the second character string is `coil`. The 11 numbers in the initialization element, including the 5th number, which codes for the coil type, are all zero. Typical examples of the of initialization and spacer `coil` element data blocks are given below.

2.1 Soft-edge magnet element block format

The first five parameters (floating-point numbers) in a soft-edge magnet element mip-file data block have a fixed meaning. Designating them by p_1 through p_5 , their meanings are:
 p_1 =magnet length L

p_2 =nominal on-axis generalized gradient G_m in teslas/meter $^{m-1}$. Exceptions: for magnet types 10 and 18, p_2 is a dimensionless scaling factor. *

p_3 =multipole index m . Read in as a floating-point number and converted to an integer. $m = 1$ for a dipole, $m = 2$ for quadrupole, etc. Negative m values are used to designate skew multipoles. For the straight-axis systems of the coil stack, $m = 1$ coils (bending magnets) are not included.

p_4 =magnet inner radius a .

p_5 =element type number *itype*, read in as a floating-point number and converted to integer **itype**.

The remaining six numbers are used for the various magnet-type elements and have type-dependent meanings.

All of the coil-stack magnet types use the magnet length $L = p_1$ in positioning the magnets in the coil array. Most of the magnet types use the magnet length explicitly in computing the gradient. The two exceptions are types that use data in input files to compute the gradient. They are types 6 and 18.

2.2 Initialization element

A typical initialization element block is

```
cinit    coil
0.000000000000000E+00  0.000000000000000E+00  0.000000000000000E+00
0.000000000000000E+00  0.000000000000000E+00  0.000000000000000E+00
0.000000000000000E+00  0.000000000000000E+00  0.000000000000000E+00
0.000000000000000E+00  0.000000000000000E+00
```

The initialization element has all zeroes, so the p_5 is zero and **itype**= 0.

2.3 Spacer element

Spacer elements are used to position coil-stack magnets in a beamline. Coil-stack spacer elements are *not* Marylie drift elements. Drift elements can be used only outside of a coil stack.

A typical spacer element block is

```
space1   coil
0.350000000000000      0.000000000000000E+00  0.000000000000000E+00
0.000000000000000E+00 -1.000000000000000E+00  0.000000000000000E+00
0.000000000000000E+00  0.000000000000000E+00  0.000000000000000E+00
0.000000000000000E+00  0.000000000000000E+00
```

Spacer elements have **itype**= -1, as in the example above. In this example, the user has given the spacer element the name **space1**. The first floating-point number is the spacer length, in this case 0.35 m. Note that negative spacers (negative p_1) can be used to superimpose magnets with different m values at a given z in the coil stack superelement. This feature can be used to add octupole correctors to quadrupole lens systems, etc.

*The magnetic scalar potential for a pure normal multipole can be written in the form $V_m(r, \theta, z) = U_m(r, z) \sin m\theta$. Then we define $g_m(z) = m \lim_{r \rightarrow 0} \frac{U_m(r, z)}{r^m}$. This definition is consistent with the usual definition of the quadrupole gradient (for which $m = 2$).

Usually, spacer elements have a fixed length in fitting loops. The most common exception is focusing of PMQ (permanent magnet quadrupole) lenses. Assuming that the PMQs have a fixed strength, the lens is focused by moving the magnets, that is, by varying the lengths of spacer elements.

3 Magnet element types

There are 17 different magnet types in the Marylie coil-stack magnet subroutines (type 9 is inactive). All of the magnets in the coil-stack menu are pure 3-D multipole magnets; if a user wishes to add higher multipole components to a particular magnet in the beamline, he can superimpose the fields of magnet elements with various m values by use of negative spacers.

Some magnet types require initialization. This is done automatically when the magnet type code indicates that a particular magnet type requires initialization, and the initialization element precedes the coil stack in the `#labor` list in the mip file. For some types, arrays of numbers are computed and stored during initialization, and used subsequently for gradient calculation. In other cases, data are read in, processed, and stored during initialization. If various magnet types do not require initialization, they are skipped during initialization. However, it is good practice to keep the `cinit` element in the `#labor` list right before a coil stack even if none of the magnet elements requires initialization.

As noted in a previous section, the first 5 numbers in a magnet element have a fixed meaning. The remaining six numbers have a type-dependent meaning.

Current-sheet magnets are described by a stream function $\psi(\phi, z)$ on a developed winding cylinder of radius a . Surface currents are given by

$$j_\phi = \frac{\partial \psi}{\partial z} \qquad j_z = -\frac{1}{a} \frac{\partial \psi}{\partial \phi} \quad (1)$$

The stream function for a normal $2m$ -pole current-sheet magnet can be expressed as a Fourier series in ϕ with z -dependent coefficients:

$$\psi(\phi, z) = \sum_{k=0}^{\infty} f_{(2k+1)m}(z) \sin[(2k+1)m]\phi \quad (2)$$

The functions $f_{(2k+1)m}(z)$ are called shape functions. When ψ contains only the fundamental ($k=0$), with either open boundaries, or with a coaxial cylinder of infinitely permeable material, the resultant field is a pure 3-D $2m$ -pole. In the open-boundary case, with the shape function $f_m(z)$, on-axis gradients and their z derivatives are computed by analytical integration over z' of a kernel $K_m(z, z')$ times the shape function $f_m(z')$. The user should note that higher multipole components due to finite turn numbers, turn-to-turn crossovers, etc. are neglected in this model.

For current-sheet magnets in open boundaries, with NI ampere-turns per pole, with the definition

$$L_{\text{eff}} = \int_{-\infty}^{+\infty} f_m(z) dz, \quad (3)$$

the z -integrated gradient is

$$(gL_{\text{eff}}) = \int_{-\infty}^{\infty} g_m(z) dz = \mu_0 NI \frac{m}{2a^m} L_{\text{eff}} \quad (4)$$

There is a current-sheet magnet type (type 17) for which the gL_{eff} of Eq. 4 is modified by a surrounding cylinder of infinitely permeable magnet material; the modified expression will be given in the description of the magnet type in the following sections. Also, some current-sheet magnets have a finite radial thickness; this is modeled by Gaussian integration of the gradients of multiple current sheets in radial depth.

Magnet gradient strengths are specified internally to Marylie by a gradient-length product computed by multiplying the nominal input gradient $G_m = p_2$ by the magnet length $L = p_1$. For most magnet types, a multiplicative factor is computed in the gradient subroutines to scale gradient profiles $g_m(z)$ such that the z -integrated gradient ($gL_{\text{eff}} = \int g_m(z)dz$) is equal to the specified gradient-length product $G_m L$. Exceptions are types 6, 10, and 18.

3.1 Type 1 magnet

This is a cylindrical current-sheet magnet in open boundaries with the shape function (centered at $z = 0$)

$$\begin{aligned} f(z) &= 1 + Az^2 + Bz^4, \quad |z| \leq L/2 \\ f(z) &= 0, \quad |z| > L/2 \end{aligned} \quad (5)$$

The coefficients A and B are given by

$$A = (s/2 - 2)/(2/L)^2 \quad B = (1 - s/2)/(2/L)^4, \quad (6)$$

where s is a dimensionless slope parameter, defined by $s = (L/2)f'(-L/2)$. A typical value for s is 1.2. The effective length of this magnet is $L_{\text{eff}} = (8 + s)L/15$. The mip-file element block for this type has the usual fixed 5 numbers followed by a single number **s** and 5 zeroes. A typical element block for this coil type with $L = 1.1$, $G = 0.2$, $m = 2$, $a = 0.778$, **itype**=1, and $s = 1.22$ follows:

```
qtype1  coil
1.1000000000000000E+00  2.0000000000000000E-01  2.0000000000000000E+00
0.7780000000000000E+00  1.0000000000000000E+00  1.2200000000000000E+00
0.0000000000000000E+00  0.0000000000000000E+00  0.0000000000000000E+00
0.0000000000000000E+00  0.0000000000000000E+00
```

Fig. 1 is a plot of the shape function for the coil described by the data block above. Fig. 2 is a contour plot of $f(z)\sin 2\phi$ for the first pole of this quadrupole winding. There are 10 contour levels: 0.05, 0.15, 0.25 $\dots$ 0.85, 0.95.

3.2 Type 2 magnet

The type 2 magnet is a radially thick Halbach-type permanent multipole magnet (PMM). The magnet model assumes a continuously varying magnetization vector of constant magnitude M_V and a rectangular cross section. The magnetization vector $\mathbf{M}$ varies with angle ϕ according to the expressions

$$\begin{aligned} M_r &= -M_V \sin m\phi \\ M_\phi &= M_V \cos m\phi \end{aligned} \quad (7)$$

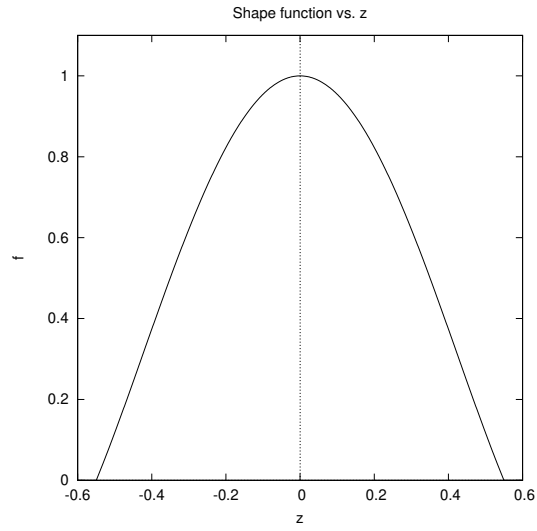


Figure 1: Shape function for type 1 magnet winding

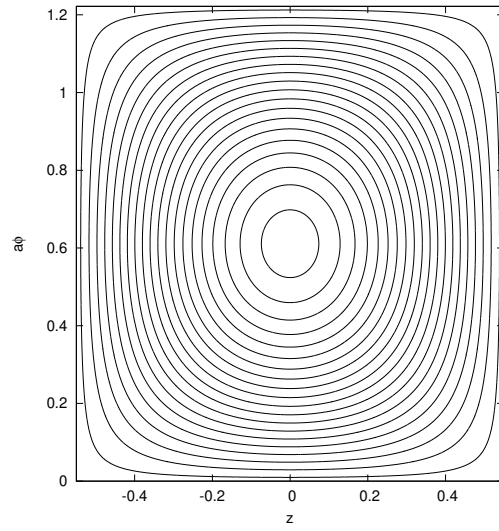


Figure 2: Winding pattern for type 1 magnet. Turn-to-turn connections are not shown.

As with current-sheet magnets, the magnet strength is specified by the gradient-length product $G_m L$, computed internally from input parameters. If we use units of teslas for M_V , M_V is approximately the remanent field of the permanent magnet material. However, since real PMMs are constructed from a finite number of wedge-shaped pieces of permanent magnet material with various constant magnetization vectors within each wedge, M_V used in the continuous model must be smaller than the remanent field of the wedges (typically by a factor of about 0.9) to take into account the finite-wedge-number effects. Then it can be shown that, for a PMM with inner radius a_1 , outer radius a_2 , and length L ,

$$\begin{aligned} (gL_{\text{eff}}) &= \int_{-\infty}^{\infty} g_m(z) dz = M_V L \ln \frac{a_2}{a_1}, \quad m = 1 \\ &= M_V L \left(\frac{1}{a_1^{m-1}} - \frac{1}{a_2^{m-1}} \right) \quad m > 1 \end{aligned} \quad (8)$$

A typical mip-file data block for a quadrupole magnet of this type is

pmq	coil		
0.0500000000000000	40.00000000000000	2.00000000000000	
0.0200000000000000	2.00000000000000	0.10000000000000	
0.0000000000000000	0.00000000000000	0.00000000000000	
0.0000000000000000	0.00000000000000		

In the data block, the first five parameters have their usual fixed meaning (note: a_1 in Eq. 8 is a); the 6th parameter is the outer radius a_2 . With these parameters, the gradient-length product is 2.0 T and the required M_V is also 1.0 T.

It should be noted that there is an alternative method in Marylie to compute transfer maps for a sequence of PMM quadrupoles ($m = 2$ only) containing higher multipoles, but with only piecewise constant sextupoles and octupoles within the quadrupoles. See Section 6.27 of the Marylie Manual, Rare Earth Cobalt (REC) Quadrupole Multiplet, Type Code: **recm**, pp. 256-274.

Charged-particle magnetic lenses can be constructed from permanent-magnet quadrupoles (PMQs). A point-to-point focusing lens with unit magnification (actually -1 magnification) in x and y can be constructed from four PMQs and field-free drifts. Since PMQs have a fixed magnetic field, fine focus adjustment is done by moving the PMQs along the beam axis. Because of the symmetry, only two distances (in three places) need to be adjusted. The following is an example of a mip file for such a lens with a fitting loop where the spaces between PMQs are adjusted for focus.

```
#comment
800 MeV P+ Russian quad identity lens
Charge= +1. Brho = 4.8810300775 Tesla-meters
Beta=0.84181072133 Gamma-1=0.8526312218 Eta=beta*gamma=1.5595648252E+00
Russian quadrupole lens. Vary quad spacing for unit magnification and focus
#beam
4.8810300775000002
0.8526312217999995
1.0000000000000000
1.0000000000000000
#menu
next      pmif
```

1.0000000000000000	17.000000000000000	2.0000000000000000
clear iden		
begin bip		
1000.0000000000000		
endfit tip		
0.0000000000000000		
robj ps9		
0.1570000000000000	0.0000000000000000	0.0000000000000000
0.0000000000000000	0.0000000000000000	0.0000000000000000
linout usr1		
3.0000000000000000	1.0000000000000000	50.000000000000000
1.0000000000000000	3.0000000000000000	0.0000000000000000
beamin usr8		
1.0000000000000000E-04	0.0000000000000000	2.4921144500000000E-03
1.0000000000000000E-04	0.0000000000000000	2.4921144500000000E-03
aim aim		
2.0000000000000000	-7.0000000000000000	0.0000000000000000
1.0000000000000000	1.0000000000000000	3.0000000000000000
vary vary		
-1.0000000000000000	-8.0000000000000000	0.0000000000000000
0.0000000000000000	1.0000000000000000	3.0000000000000000
fit fit		
10.000000000000000	0.0000000000000000	0.0000000000000000
1.0000000000000000E-02	0.0000000000000000	3.0000000000000000
cinit coil		
0.0000000000000000E+00	0.0000000000000000E+00	0.0000000000000000E+00
0.0000000000000000E+00	0.0000000000000000E+00	0.0000000000000000E+00
0.0000000000000000E+00	0.0000000000000000E+00	0.0000000000000000E+00
0.0000000000000000E+00	0.0000000000000000E+00	0.0000000000000000E+00
indrft coil		
2.0000000000000000	0.0000000000000000E+00	0.0000000000000000E+00
0.0000000000000000E+00	-1.0000000000000000	0.0000000000000000E+00
0.0000000000000000E+00	0.0000000000000000E+00	0.0000000000000000E+00
0.0000000000000000E+00	0.0000000000000000E+00	0.0000000000000000E+00
endrft coil		
2.0000000000000000	0.0000000000000000E+00	0.0000000000000000E+00
0.0000000000000000E+00	-1.0000000000000000	0.0000000000000000E+00
0.0000000000000000E+00	0.0000000000000000E+00	0.0000000000000000E+00
0.0000000000000000E+00	0.0000000000000000E+00	0.0000000000000000E+00
espace coil		
0.3000000000000000	0.0000000000000000E+00	0.0000000000000000E+00
0.0000000000000000E+00	-1.0000000000000000	0.0000000000000000E+00
0.0000000000000000E+00	0.0000000000000000E+00	0.0000000000000000E+00
0.0000000000000000E+00	0.0000000000000000E+00	0.0000000000000000E+00
mspace coil		
0.3000000000000000	0.0000000000000000E+00	0.0000000000000000E+00

```

0.0000000000000000E+00 -1.0000000000000000 0.0000000000000000E+00
0.0000000000000000E+00 0.0000000000000000E+00 0.0000000000000000E+00
0.0000000000000000E+00 0.0000000000000000E+00
pmqa coil
0.1500000000000000 30.000000000000000 2.000000000000000
0.0400000000000000 2.000000000000000 0.100000000000000
0.0000000000000000 0.000000000000000 0.000000000000000
0.0000000000000000 0.000000000000000
pmqb coil
0.3000000000000000 -30.000000000000000 2.000000000000000
0.0400000000000000 2.000000000000000 0.100000000000000
0.0000000000000000 0.000000000000000 0.000000000000000
0.0000000000000000 0.000000000000000
pmqc coil
0.3000000000000000 30.000000000000000 2.000000000000000
0.0400000000000000 2.000000000000000 0.100000000000000
0.0000000000000000 0.000000000000000 0.000000000000000
0.0000000000000000 0.000000000000000
pmqd coil
0.1500000000000000 -30.000000000000000 2.000000000000000
0.0400000000000000 2.000000000000000 0.100000000000000
0.0000000000000000 0.000000000000000 0.000000000000000
0.0000000000000000 0.000000000000000
intg intg
6.000000000000000 1.000000000000000 0.000000000000000E+00
1.000000000000000 2000.000000000000 36.0000000000000
mapout ptm
3.000000000000000 3.000000000000000 3.000000000000000
3.000000000000000 1.000000000000000
mapsav tmo
16.000000000000000
magmap tmo
56.000000000000000
store stm
2.000000000000000
getmap gtm
1.000000000000000 2.000000000000000
taylor3 usr2
1.000000000000000 4.000000000000000 99.0000000000000
0.000000000000000 0.000000000000000 0.000000000000000
fin end
#xdata
ixd>: 7
R(1,1)=-1.0 R(1,2)=0.0 #
ixd>: 8
espace 1 mspace 1 #

```

```

#
#lines
  cident
    1*indrft      1*pmqa      1*espace      1*pmqb      1*mspace  &
    1*pmqc      1*espace      1*pmqd      1*endrft
#lumps
#loops
#labor
  1*begin
  1*clear
  1*cinit
  1*cident
  1*intg
  1*aim
  1*vary
  1*fit
  1*endfit
  1*clear
  1*cinit
  1*cident
  1*intg
  1*magmap
  1*mapout
  1*mapsav
  1*next
  1*fin

```

The quadrupole spacings that are varied are `espace` and `mspace`. Initial values are for both are 0.3 m; after focus they are 0.198098469646549 m and 0.320997754750815 m, respectively.

Figure 3 shows the gradient plot after focusing; it can be seen that the fringe fields of `pmqa` and `pmqb` and also `pmqc` and `pmqd` overlap considerably, which reduces their effective gradient-length products.

3.3 Type 3 and type 4 magnets

Coil types 3 and 4 (`itype=3` or `itype=4` in the coil element block) represent the fundamental and first “allowed” harmonic gradient components produced by a single Lambertson-type printed-circuit coil with open boundaries. Lambertson coils do not produce a pure 3-D multipole field; the field (scalar potential) is $\sin m\theta$ only in the z -integrated sense. In fact, the field produced by a Lambertson coil has strong $3m$ and higher components that integrate to zero in z . The m and $3m$ components are computed by the same subroutine in Marylie, but in separate calls to the subroutine with either `itype=3` or `itype=4`. The m and $3m$ gradient components for a particular Lambertson coil are computed by a block of three coil elements, consisting of a coil element with `itype=3`, followed by a negative spacer element of length $L = p_1$, followed by a coil element with `itype=4`. In order to ensure that the m and $3m$ components of a particular coil are produced by the same windings and same current, the numbers in the element blocks for both `itype=3` and

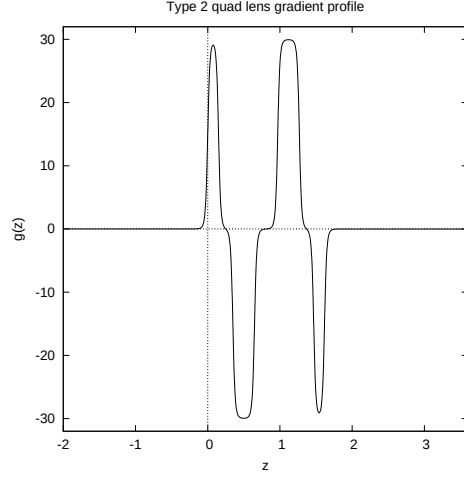


Figure 3: Gradient profile for the unit magnification lens after focusing

itype=4 are the same except that **itype=3** has $p_3 = m$ and **itype=4** has $p_3 = 3m$. The negative spacer causes the $3m$ component to be centered on the m component.

The windings of a Lambertson coil consist of turn segments parallel to the axis of the winding cylinder that are connected by crossover turns perpendicular to the axis. With the stipulation that the end crossover turns are equally spaced, the winding law is

$$l(\phi) = \sqrt{L^2 - (L^2 - L_{\min}^2) \sin^2 m\phi} \quad (9)$$

where $l(\phi)$ is the length of the turn segments parallel to the axis, L is the coil length, and $L_{\min}$ is the limiting length of the parallel turns at the pole center. In some Lambertson coils, $L_{\min}$ is zero. Fig. 4 is a filament model of the windings that was made by plotting equally spaced contours of the stream function.

The shape function profiles for the various multipole components of the open-boundary field generated by the Lambertson coil are found by Fourier-analyzing the stream function in angle. This is done analytically. In practice, only the fundamental m and the first “allowed” harmonic $3m$ are needed.

Since the coil stack presently applies only to straight-axis systems, multipole numbers m only of two and higher are treated. In the case of $m = 1$ (steering magnets), the bending effect cannot be modeled and coil elements with **itype=3** and $m = 1$ are not allowed. However, the sextupole component of a $m = 1$ coil can be treated with a single element block with **itype=4** and $m = 3$. The p_2 value in this case is the nominal bending field of the steering magnet. In a 3rd-order code, the first “allowed” field component of quadrupole Lambertson magnets is $m = 6$ which contributes only to 5th order terms in the map. Therefore an **itype=4** element has no effect on the map and can be omitted. With a fifth-order code, $m = 2$ and the first “allowed” harmonic $m = 6$ would

both contribute to the transfer map.

A typical element block for the sextupole component of a Lambertson steering magnet with a nominal bending field of 0.1 T follows:

```
steer4  coil
0.300000000000000000 0.1 3.0000000000000000
0.060000000000000000 4.0000000000000000 0.1000000000000000
0.000000000000000000 0.0000000000000000 0.0000000000000000
0.000000000000000000 0.0000000000000000
```

The first five numbers in the block have their usual meanings; the sixth number is $L_{\min}$. A typical set of three element blocks for a quadrupole Lambertson magnet, including a negative drift and a duodecapole component follows. These three elements would be used for a fifth-order version of Marylie; if used in a third-order version of Marylie the combined `nspac` and `lamb4` elements would have no effect on the map.

```
lamb3      coil
0.300000000000000000 1.117 2.0000000000000000
0.060000000000000000 3.0000000000000000 0.1000000000000000
0.000000000000000000 0.0000000000000000 0.0000000000000000
0.000000000000000000 0.0000000000000000

nspac      coil
-0.300000000000000000 0.0000000000000000 0.0000000000000000
0.000000000000000000 -1.0000000000000000 0.0000000000000000
0.000000000000000000 0.0000000000000000 0.0000000000000000
0.000000000000000000 0.0000000000000000

lamb4      coil
0.300000000000000000 1.117 6.0000000000000000
0.060000000000000000 4.0000000000000000 0.1000000000000000
0.000000000000000000 0.0000000000000000 0.0000000000000000
0.000000000000000000 0.0000000000000000
```

3.4 Type 5 magnet

This magnet has a square-edged shape function in open boundaries. That is, $f_m(z) = 1$, $-L/2 < z < L/2$; $f_m(z) = 0$, $|z| > L/2$. Accordingly, the current density is (unphysically) infinite where the crossover segments turns at the ends are superimposed. This magnet type is typically used to represent higher-order correctors. The effective length is L . A typical element block for an octupole coil of this type follows:

```
otype5  coil
0.5  80.0  4.0
0.01  5.0  0.0
0.0  0.0  0.0
0.0  0.0
```

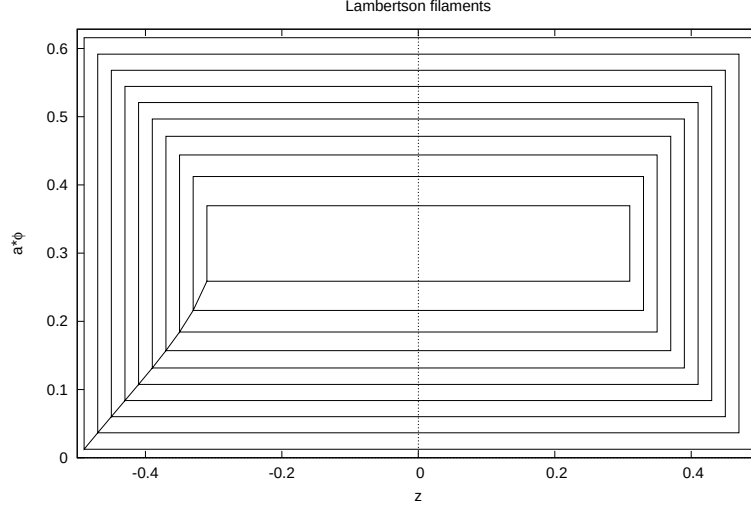


Figure 4: Ten-turn filament model for a Lambertson-type current-sheet magnet with $L = 1.0$, $L_{\min} = 0.6$, $a = 0.2$, and $m = 1$. Only one pole is shown. The spiral nature of physically realistic windings with turn-to-turn connections is neglected.

3.5 Type 6 magnet

This magnet type has a user-defined shape function defined by a set of points (z, f) read in from a two-column file. The points are read in and converted to a piecewise-continuous quadratic interpolant during an initialization call. The data file is read in with the default Fortran file name **fort.ifile**, where ifile is an integer that is the 6th number in the coil data block. The z values in the data file range monotonically from $-L/2$ to $L/2$. The first (z, f) value in the data file should be $(-L/2, 0)$ and the last (z, f) value should be $(L/2, 0)$.

Since the user-defined shape function can have any profile, including zones with negative values, and possibly integrate in z to zero, the the strength constant used in computing the gradient is not based on $p_1 p_2 / L_{\text{eff}}$, but just p_2 (G_m).

A typical element block for a quadrupole type 6 magnet with $L = 3.1$, $G_m = 27.657$, $a = 0.2$, and **ifile**=155 follows:

```
qtype6    coil
3.1000000000000000    27.657    2.0000000000000000
0.2000000000000000    6.0000000000000000    155.00000000000000
0.0000000000000000    0.0000000000000000    0.0000000000000000
0.0000000000000000    0.0000000000000000
```

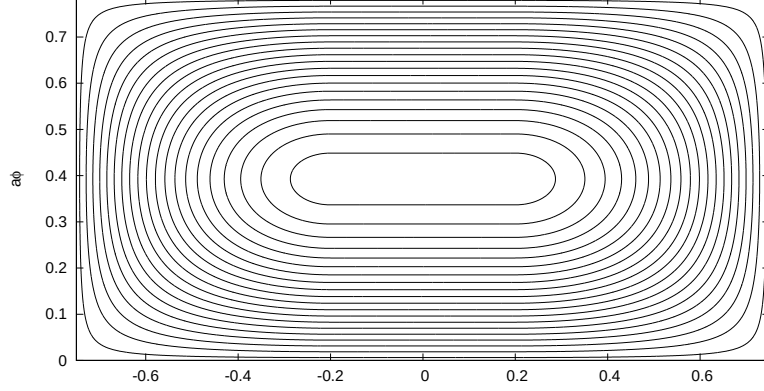


Figure 5: Ten-turn filament model for a type 7 current-sheet magnet with $L = 1.5$, $m = 2$, $a = 0.5$, and $w_{\text{flat}} = 0.4$. Only one pole is shown. The spiral nature of physically realistic windings with turn-to-turn connections is neglected.

3.6 Type 7 magnet

The type 7 magnet is a current-sheet magnet with a shape function that has a flat region in the center of width w_{flat} and end zones with a quadratic z dependence:

$$\begin{aligned}
 f(z) &= 1 - \frac{[w_{\text{flat}}/2 + z]^2}{[L/2 - w_{\text{flat}}/2]^2}, & -L/2 \leq z \leq -w_{\text{flat}}/2 \\
 f(z) &= 1, & -w_{\text{flat}}/2 \leq z \leq w_{\text{flat}}/2 \\
 f(z) &= 1 - \frac{[w_{\text{flat}}/2 - z]^2}{[L/2 - w_{\text{flat}}/2]^2}, & w_{\text{flat}}/2 \leq z \leq L/2 \\
 f(z) &= 0, & |z| > L/2
 \end{aligned} \tag{10}$$

The effective length for this magnet type is $L_{\text{eff}} = \frac{2}{3}L + \frac{1}{3}w_{\text{flat}}$. A typical mip-file data block for a quadrupole magnet of this type with $L = 1.5$, $G_m = 2.0$, $m = 2$, $a = 0.5$, **itype=7**, and $w_{\text{flat}} = 0.4$ is

```

qtype7  coil
1.5000000000000E+00  2.0000000000000E+00  2.0000000000000E+00
0.5000000000000E+00  7.0000000000000E+00  0.4000000000000E+00
0.0000000000000E+00  0.0000000000000E+00  0.0000000000000E+00
0.0000000000000E+00  0.0000000000000E+00

```

Fig. 5 is a filament model of the windings that was made by plotting equally spaced contours of the stream function.

3.7 Type 8 magnet

The type 8 magnet is a radially thin Halbach-type permanent multipole magnet (PMM). This magnet type was originally used to model large-bore, low-gradient permanent-magnet quadrupole

magnets made by mounting many rows of equal-strength permanent-magnet blocks parallel to the magnet axis on a circular support structure. The directions of the magnetization vectors of the rows were varied to give the same rotation as with the type 2 magnets: $M_r = -M_S \sin m\phi$, $M_\phi = M_S \cos m\phi$, with $m = 2$. Block thicknesses are small compared to the magnet mean radius. Again, finite block effects are neglected and the magnet model used for gradient calculation assumes a continuously varying magnetization vector of constant magnitude M_S . M_S is a average over the circumference and less than M_V of the individual blocks times the radial depth of the blocks. As with current-sheet magnets, the magnet strength is specified by the gradient-length product $G_m L$. Since M_s here is a surface, not a volume magnetization density, it has units of tesla-meters. Then it can be shown that, for a thin PMM with mean radius a and length L ,

$$(gL_{\text{eff}}) = \int_{-\infty}^{\infty} g_m(z) dz = \frac{mM_s}{a^m} L \quad (11)$$

A typical mip-file data block for a quadrupole magnet of this type is

thnmpmq	coil		
1.0000000000000000	0.1000000000000000	2.0000000000000000	
0.5000000000000000	8.0000000000000000	0.0000000000000000	
0.0000000000000000	0.0000000000000000	0.0000000000000000	
0.0000000000000000	0.0000000000000000		

With these parameters, the gradient-length product is 0.1 T and the required M_s is 0.0125 T-m.

3.8 Type 9 magnet

At present there is no type 9 magnet.

3.9 Type 10 magnet

This magnet type is a permanent multipole magnet of rectangular cross section with a volume dipole density in the z direction varying as $\sin m\phi$. This magnet type is not commonly used, but has interesting properties. A magnet of this type could be made with wedge-shaped permanent magnet pieces of constant magnetization with appropriate gaps to approximate a $\sin m\phi$ angular dependence. The on-axis gradient integrates to zero and for $m = 2$ has the shape of the gradient of a closely-spaced quadrupole doublet. The first 5 parameters in the mip file for this magnet are the usual fixed-meaning parameters length L , nominal gradient G_m , multipole number m , inner radius a , and type code **itype**. The additional parameters are volume magnetization B_{rem} (units of teslas) and outer radius a_2 . The expression for the on-axis gradient for this magnet type with $a_1 = a$ is

$$g_m(z) = B_{\text{rem}} \frac{(2m-1)!!}{2^{m+1}(m-1)!} \int_{a_1}^{a_2} \rho^{m+1} \left\{ \frac{1}{[\rho^2 + (z - L/2)^2]^{m+1/2}} - \frac{1}{[\rho^2 + (z + L/2)^2]^{m+1/2}} \right\} d\rho \quad (12)$$

The integral in Eq. 12 is evaluated numerically with Gaussian integration. The integrated gradient for a type 10 magnet is zero. Therefore the gradient parameter p_2 is used differently. For this type, it is dimensionless and has nominal value 1. It can be used as a fitting parameter to scale the

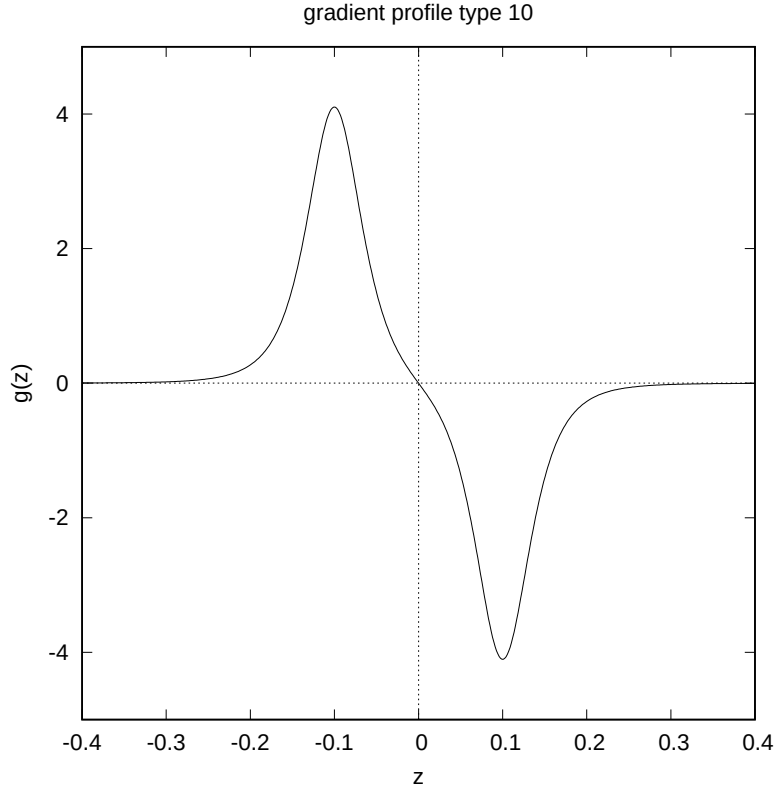


Figure 6: Plot of the gradient of a type 10 magnet *vs.* z

on-axis gradient values.

A typical mip-file data block for a quadrupole magnet of this type with $L = 0.2$, $G_m = 1$, $m = 2$, $a = 0.05$, `itype=10`, $B_{\text{rem}} = 1.1$, and $a_2 = 0.1$ is

pmqz	coil	
0.2000000000000000	1.0000000000000000	2.0000000000000000
5.000000000000000E-02	10.000000000000000	1.1000000000000000
0.1000000000000000	0.0000000000000000	0.0000000000000000
0.0000000000000000	0.0000000000000000	

Figure 6 is a plot of the gradient for the parameters of the mip-file data block.

3.10 Type 11 magnet

The type 11 magnet is a cylindrical surface winding without iron, with a shape function that is flat in the center and rounded at the ends. The ends are quartic in distance from flattop. Besides the usual fixed five parameters L , G_m , m , a , and `itype`, there are four additional parameters: w_1 , s_1 , w_2 , and s_2 . w_1 and w_2 are end widths and s_1 and s_2 are dimensionless slopes, with $s_1 = w_1 f'(-L/2)$

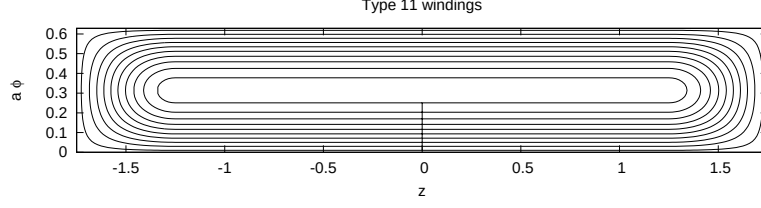


Figure 7: Winding pattern for a type 11 magnet with the parameters of the type 11 mip file data block. Turn-to-turn connections are not shown.

and $s_2 = -w_2 f'(L/2)$. The shape function is defined to be

$$\begin{aligned}
 f(x) &= 1 + A_1(x + L/2 - w_1)^2 + B_1(x + L/2 - w_1)^4, & -L/2 < x < -L/2 + w_1 \\
 &= 1, & -L/2 + w_1 < x < L/2 - w_2 \\
 &= 1 + A_2(x - L/2 + w_2)^2 + B_2(x - L/2 + w_2)^4, & L/2 - w_2 < x < L/2
 \end{aligned} \quad (13)$$

where

$$\begin{aligned}
 A_1 &= (s_1/2 - 2)/w_1^2 & B_1 &= (1 - s_1/2)/w_1^4 \\
 A_2 &= (s_2/2 - 2)/w_2^2 & B_2 &= (1 - s_2/2)/w_2^4
 \end{aligned} \quad (14)$$

A typical mip-file data block for a quadrupole magnet of this type with $L = 3.5$, $G_m = 17.0$, $m = 2$, $a = 0.2$, `itype`=11, $w_1 = 0.5$, $w_2 = 0.6$, $s_1 = 1.0$, and $s_2 = 1.1$ is

```

qtyp11 coil
3.5000000000000000    17.000000000000000    2.0000000000000000
0.2000000000000000    11.000000000000000    0.5000000000000000
0.6000000000000000    1.0000000000000000    1.1000000000000000
0.0000000000000000    0.0000000000000000

```

This data block produces windings that are different in the two ends. Figure 7 is a filament model of the type11 windings with the parameters of the type 11 element data block shown above.

3.11 Type 12 magnet

The type 12 magnet is a cylindrical surface winding without iron, with a shape function that is flat in the center and rounded at the ends. The ends are cubic in distance from flattop. Besides the usual fixed five parameters L , G_m , m , a , and `itype`, there are four additional parameters: w_1 , s_1 , w_2 , and s_2 . w_1 and w_2 are end widths and s_1 and s_2 are dimensionless slopes, defined in the same way as those of the type 11 magnet. The shape function is defined to be

$$\begin{aligned}
 f(x) &= 1 + A_1(x + L/2 - w_1)^2 - B_1(x + L/2 - w_1)^3, & -L/2 < x < -L/2 + w_1 \\
 &= 1, & -L/2 + w_1 < x < L/2 - w_2 \\
 &= 1 + A_2(x - L/2 + w_2)^2 + B_2(x - L/2 + w_2)^3, & L/2 - w_2 < x < L/2
 \end{aligned} \quad (15)$$

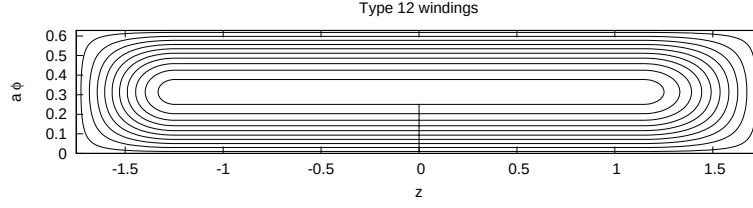


Figure 8: Winding pattern for a type 12 magnet with the parameters of the type 12 mip file data block. Turn-to-turn connections are not shown.

where

$$\begin{aligned} A_1 &= (s_1 - 3)/w_1^2 & B_1 &= (2 - s_1)/w_1^3 \\ A_2 &= (s_2 - 3)/w_2^2 & B_2 &= (2 - s_2)/w_2^3 \end{aligned} \quad (16)$$

A typical mip-file data block for a quadrupole magnet of this type with $L = 3.5$, $G_m = 17.0$, $m = 2$, $a = 0.2$, `itype=12`, $w_1 = 0.5$, $w_2 = 0.6$, $s_1 = 1.0$, and $s_2 = 1.1$ is

```
qtyp12 coil
3.5000000000000000    17.000000000000000    2.0000000000000000
0.2000000000000000    12.000000000000000    0.5000000000000000
0.6000000000000000    1.0000000000000000    1.1000000000000000
0.0000000000000000    0.0000000000000000
```

This data block produces windings that are different in the two ends. Figure 8 is a filament model of the type12 windings with the parameters of the type 12 element data block shown above.

3.12 Type 13 magnet

This magnet has the same shape function as the type 1 magnet, but has finite radial thickness. As with all of the radially thick electromagnets magnets in the coil menu, the ox-axis gradient is computed by Gaussian numerical integration in coil radius.

The mip-file element block for this type has the usual fixed 5 numbers followed two more numbers: s and a_2 , where a_2 is the outer radius. The rest of the block is padded out with 4 zeroes. A typical element block for this coil type follows:

```
qtyp13 coil
1.1000000000000000E+00  5.000000000000000E-02  2.000000000000000E+00
0.7780000000000000E+00  13.000000000000000E+00  1.220000000000000E+00
0.9000000000000000E+00  0.000000000000000E+00  0.000000000000000E+00
0.0000000000000000E+00  0.000000000000000E+00
```

As an alternative to use of this and other radially thick coil types, the user may want to use negative drifts to superimpose the gradients of several thin coils of increasing radius. This alternative allows more flexibility, since end zone dimensions can change with coil radius.

3.13 Type 14 magnet

The type 14 magnet has the same shape function as the type 7 magnet, but is radially thick with outer radius a_2 . A typical mip-file data block for a quadrupole magnet of this type with $L = 1.5$, $G_m = 2.0$, $m = 2$, $a = 0.5$, `itype=14`, $w_{\text{flat}} = 0.4$, and $a_2 = 0.6$ is

```
qtyp14  coil
1.5000000000000000E+00  2.0000000000000000E+00  2.0000000000000000E+00
0.5000000000000000E+00  14.0000000000000000E+00  0.4000000000000000E+00
0.6000000000000000E+00  0.0000000000000000E+00  0.0000000000000000E+00
0.0000000000000000E+00  0.0000000000000000E+00
```

3.14 Type 15 magnet

The type 15 magnet has the same shape function as the type 11 magnet, but is radially thick with outer radius a_2 . A typical mip-file data block for a quadrupole magnet of this type with $L = 3.5$, $G_m = 17.0$, $m = 2$, $a = 0.4$, `itype=15`, $w_1 = 0.5$, $w_2 = 0.6$, $s_1 = 1.0$, $s_2 = 1.1$, and $a_2 = 0.5$ is

```
qtyp15 coil
3.5000000000000000      17.000000000000000      2.0000000000000000
0.4000000000000000      15.000000000000000      0.5000000000000000
0.6000000000000000      1.0000000000000000      1.1000000000000000
0.5000000000000000      0.0000000000000000
```

3.15 Type 16 magnet

The type 16 magnet has the same shape function as the type 12 magnet, but is radially thick with outer radius a_2 . A typical mip-file data block for a quadrupole magnet of this type with $L = 3.5$, $G_m = 17.0$, $m = 2$, $a = 0.2$, `itype=16`, $w_1 = 0.5$, $w_2 = 0.6$, $s_1 = 1.0$, $s_2 = 1.1$ and $a_2 = 0.25$ is

```
qtyp16 coil
3.5000000000000000      17.000000000000000      2.0000000000000000
0.2000000000000000      16.000000000000000      0.5000000000000000
0.6000000000000000      1.0000000000000000      1.1000000000000000
0.2500000000000000      0.0000000000000000
```

3.16 Type 17 magnet

The type 17 magnet has the same shape function as the type 11 magnet, but is radially thick with outer radius a_2 and has a surrounding concentric iron surface of radius b . The effect of the surrounding iron cylinder is computed in the infinite permeability limit. The iron surface has two effects: it increases the gradient-length product (z -integrated gradient) and causes the magnetic field and on-axis gradient to fall off exponentially in z past the end of the windings (see LA-UR-90-4090, Eqs. 99 and 100). By contrast, the gradients for all of the current-dominated magnets with open boundaries fall off with the power law $g \propto 1/|z|^{2m+1}$ for large z .

The expression for the gradient-length product of a surface-current magnet with the surrounding iron cylinder is the same as that of the open-boundary case (Eq. 4) except for an additional factor:

$$(gL_{\text{eff}}) = \mu_0 N I \frac{m}{2a^m} L_{\text{eff}} \left[1 + \left(\frac{a}{b} \right)^{2m} \right] \quad (17)$$

The computed gradient is the sum of two terms: a term that is the same as the open-boundary gradient of the bare coil, plus a contribution from the iron surface. The iron contribution retains the pure 3-D multipole $\sin m\theta$ symmetry of the bare windings. The iron-surface gradient term is computed from an equivalent fictitious surface current at radius b represented by a shape function. Since the type 17 magnet has radially thick windings, the iron shape function is computed by Gaussian numerical integration in coil radius of the contributions of multiple current-sheet magnets of increasing radius.

The shape function for the iron contribution is computed at a set of z points and stored during an initialization step. The iron shape function is computed by a numerical Fourier integral subroutine based on modified Bessel functions.

The exponential falloff of the gradient results numerically from the near cancellation of the bare-coil and iron gradient terms at large z . A typical mip-file data block for a type 17 quadrupole magnet with $L = 3.1$, $G_m = 17.129$, $m = 2$, $a = 0.182$, `itype=17`, $w_1 = 0.3$, $w_2 = 0.3$, $s_1 = 1.0$, $s_2 = 1.0$, $a_2 = 0.194$ and $b = 0.26$ is

<code>qtyp17</code>	<code>coil</code>		
3.100000000000000	17.1290000000000	2.000000000000000	
0.182000000000000	17.0000000000000	0.300000000000000	
0.300000000000000	1.000000000000000	1.000000000000000	
0.194000000000000	0.260000000000000		

The gradient profile for the above example is shown in Fig. 9.

3.17 Type 18 magnet

This magnet type provides users a means of exactly modeling the gradient profile for magnets of any kind based on measured magnetic field data on a reference cylinder passing through the bore of the magnet. The user provides an input file containing coefficients for cubic interpolants in z of angular Fourier components of the field components B_r and B_ϕ . The user must process measured or computed field data on the reference cylinder of radius r_{ref} to prepare the input data file. The Fourier coefficients for the normal field components have the usual definitions

$$b_r^m(z) = \frac{1}{\pi} \int_0^{2\pi} B_r(r_{\text{ref}}, \phi, z) \sin m\phi d\phi \quad b_\phi^m(z) = \frac{1}{\pi} \int_0^{2\pi} B_\phi(r_{\text{ref}}, \phi, z) \cos m\phi d\phi \quad (18)$$

Skew Fourier coefficients are evaluated in an analogous way. The integrals in Eq. 18 are evaluated numerically using magnet data. After $b_r^m(z)$ and $b_\phi^m(z)$ coefficients on a set of z values are obtained, the user next must find a piecewise cubic representation in z for the $b_r^m(z)$ and $b_\phi^m(z)$ data sets. They could be obtained from an interpolating cubic spline with nodes at all of the z values in the data set, or by use of a least-squares-fit spline based on cubic B-splines, with multiple data points between knots. In the first case, the user data file retains all of the z values of the field data set, while in the second case there are fewer z values in the user data file than in the field data set. In

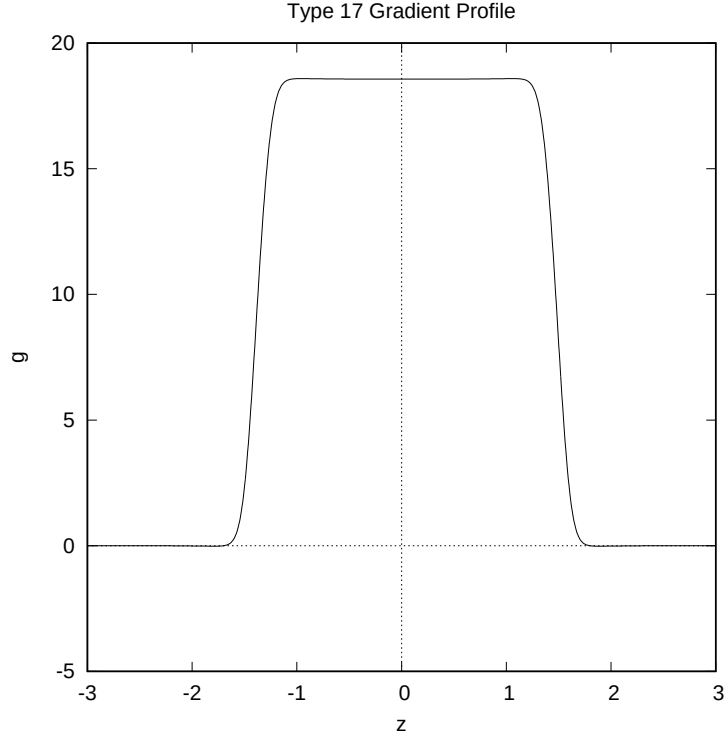


Figure 9: Gradient profile for a type 17 quadrupole

either case, with $N - 1$ z -intervals $[z_1, z_2], [z_2, z_3], \dots, [z_n, z_{n+1}], \dots, [z_{N-1}, z_N]$ we define the cubic coefficients for the user input data set on the interval $[z_n, z_{n+1}]$ by

$$\begin{aligned} f_r^m(z) &= a_1 + b_1(z - z_n) + c_1(z - z_n)^2 + d_1(z - z_n)^3 \\ f_\phi^m(z) &= a_2 + b_2(z - z_n) + c_2(z - z_n)^2 + d_2(z - z_n)^3 \end{aligned} \quad (19)$$

where $f_r^m(z)$ is the cubic approximation function for $b_r^m(z)$ and $f_\phi^m(z)$ is the cubic approximation function for $b_\phi^m(z)$.

Fourier components for various m values in a particular user magnet are treated as separate coils in a coil stack and the appropriate spacer stack elements are used to superimpose the various Fourier components.

The user data file is read in with the default Fortran input file name of the form `fort.ifile`, where ifile is an integer that is the 6th number in the coil data block. Since other files with a default Fortran file name of the form `fort.ifile` are frequently used in Marylie runs, the user must be careful to chose a value for ifile in `fort.ifile` that does not conflict with them.

A typical mip-file data block for a type 18 magnet (a quadrupole in this case) with $L = 4.0, G_m = 2.84700434343684, m = 2, a = 0.1, itype=18$, and `ifile=98` is

aqu	coil		
4.000000000000000		2.84700434343684	2.000000000000000
0.100000000000000		18.0000000000000	98.0000000000000
0.000000000000000		0.000000000000000	0.000000000000000
0.000000000000000		0.000000000000000	

For type 18 coils, $G_m(p_2)$ is a dimensionless scaling factor that multiplies the gradient computed from the user data file data. In this case, the gradient computed for the type 18 magnet is that computed from the user data file, multiplied by 2.84700434343684. The nominal coil radius a is 0.1. The input a is not actually used in gradient calculation; the reference radius in the user data file is used instead; this will be explained below. The value `ifile=98` indicates that during initialization of the above coil the user data will be read in from the file `fort.98`. A mip file for a four-quadrupole focusing lens using the type 18 quadrupole above follows. Note the use of the negative spacer element `nspc` to bring the central two quadrupoles closer together. Alternatively, the nominal lengths of the quadrupoles could have been made smaller.

```
#comment
This is kinematics data for the 50 Gev reference accelerator
Mass = 938.2796   K.E.= 50000.0000 Mev   P= 50929.6373 Mev/c
Charge= 1.        Brho = 169.88298399 Tesla-meters
Beta (v/c)= 0.99983034   Gamma= 54.28901961   Eta= 54.27980886

#beam
169.88298399449701
53.289019605669800
1.000000000000000
1.000000000000000

#menu
next      pmif
1.000000000000000    17.0000000000000    2.000000000000000
clear     iden
begin     bip
1000.0000000000000
cinit     coil
0.000000000000000    0.000000000000000    0.000000000000000
0.000000000000000    0.000000000000000    0.000000000000000
0.000000000000000    0.000000000000000    0.000000000000000
0.000000000000000    0.000000000000000
aqu       coil
4.000000000000000    2.84700434343684    2.000000000000000
0.100000000000000    18.0000000000000    98.0000000000000
0.000000000000000    0.000000000000000    0.000000000000000
0.000000000000000    0.000000000000000
bqu       coil
4.000000000000000    -2.84700434343684    2.000000000000000
0.100000000000000    18.0000000000000    98.0000000000000
0.000000000000000    0.000000000000000    0.000000000000000
0.194000000000000    0.000000000000000
```

```

intg      intg
  6.000000000000000      1.000000000000000      0.000000000000000
  1.000000000000000      2000.0000000000000      36.000000000000000
nspace    coil
-0.400000000000000      0.000000000000000      0.000000000000000
  0.000000000000000     -1.000000000000000      0.000000000000000
  0.000000000000000      0.000000000000000      0.000000000000000
  0.000000000000000      0.000000000000000
focu      coil
  2.550000000000000      0.000000000000000      0.000000000000000
  0.000000000000000     -1.000000000000000      0.000000000000000
  0.000000000000000      0.000000000000000      0.000000000000000
  0.000000000000000      0.000000000000000
mapout    ptm
  3.000000000000000      3.000000000000000      3.000000000000000
  3.000000000000000      1.000000000000000
mapsav    tmo
  16.000000000000000
magmap    tmo
  56.000000000000000
square    sqr
fin        end
#lines
cell
  1*focu      1*aqu      1*nspace      1*bqu      1*focu
#lumps
#loops
#labor
  1*begin
  1*clear
  1*cinit
  1*cell
  1*intg
  1*magmap
  1*square
  1*mapout
  1*fin

```

3.17.1 Type 18 User data input file format

The data file starts with a one-line header containing the variables **m** (multipole number) and **rref** (field data reference radius), followed by any alphanumeric character strings that the user may add for file identification purposes. Only the numbers **m**, **rref** in the header are used for gradient computation. The **m** value in the header line and the **m** in the element data block ($p(3)$) are checked for compatibility: they must be the same.

The rest of the user data file consists of pairs of lines containing the starting z of intervals

followed by the cubic interpolation coefficients on the intervals (see Eq. 19) in the order

z a_1 b_1 c_1 d_1

a_2 b_2 c_2 d_2

All numbers are in exponential format with a maximum width up to 20, e.g. **format 1pe20.13**. The z values are usually distributed around the center of the magnet with $z = 0$ in the middle of the lines. There is one more z value in the data set than there are intervals. The first interval is $[z_1, z_2]$, etc. The 8 coefficients following the last z value in the data file (the last two lines) are dummies and can be zeroes or repeats of the coefficients of the previous two lines. The header, lines for the first two intervals, and lines for the last interval of a typical user data file with the first z interval $[-2.000000000000E+00, -1.995000000000E+00]$ and the last z interval $[1.995000000000E+00, 2.000000000000E+00]$ follow:

```

2 8.0000000000E-02 1000 lambin2.dat          m rref ntheta ifile
-2.000000000000E+00 -2.811133526000E-06 -7.0558920248789E-06 0.000000000000E+00 -9.0277420484907E-04
-2.8224756226000E-06 -7.0959039599776E-06 0.000000000000E+00 -9.0963680090273E-04
-1.9950000000000E+00 -2.8465258329000E-06 -7.1236000902426E-06 -1.3541613072736E-05 2.2591662425654E-04
-2.8580688470000E-06 -7.1641267200453E-06 -1.3644552013541E-05 2.2759920452546E-04
. . . . .
1.9950000000000E+00 -2.8465258329000E-06 7.1236000809605E-06 -1.3541616288112E-05 9.0277441920747E-04
-2.8580688471000E-06 7.1641267052183E-06 -1.3644547565451E-05 9.0963650436344E-04
2.0000000000000E+00 -2.8465258329000E-06 7.1236000809605E-06 -1.3541616288112E-05 9.0277441920747E-04
-2.8580688471000E-06 7.1641267052183E-06 -1.3644547565451E-05 9.0963650436344E-04

```

4 References

[1] P. L. Walstrom, “Magnetic Fields and Inductances of Cylindrical Current Sheet Magnets”, Los Alamos National Laboratory Report LA-UR-90-4090, 1990.